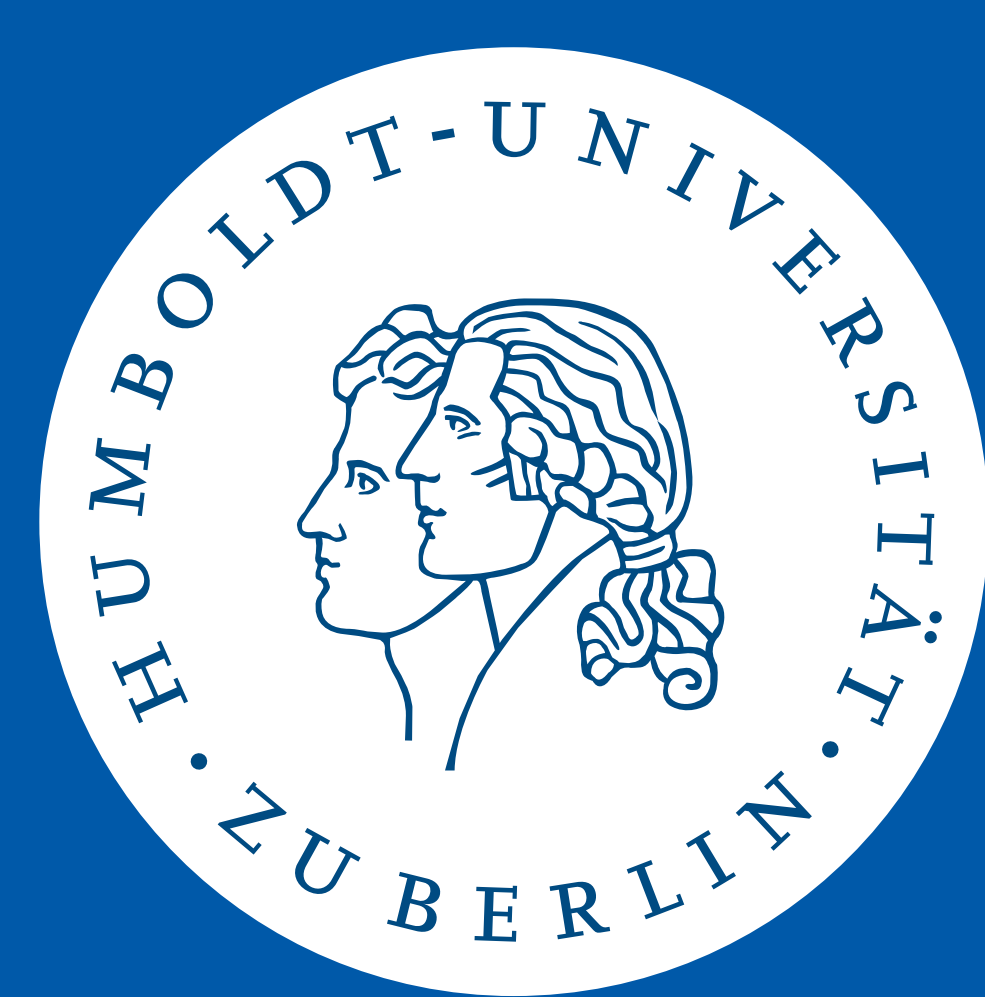
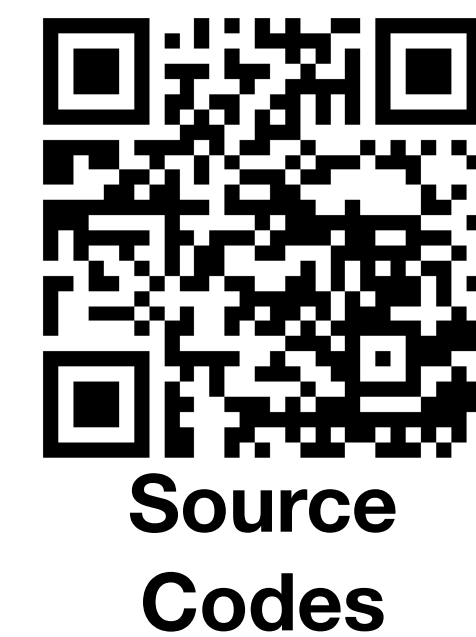


Discovering Leitmotifs in Multidimensional Time Series



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Leitmotifs

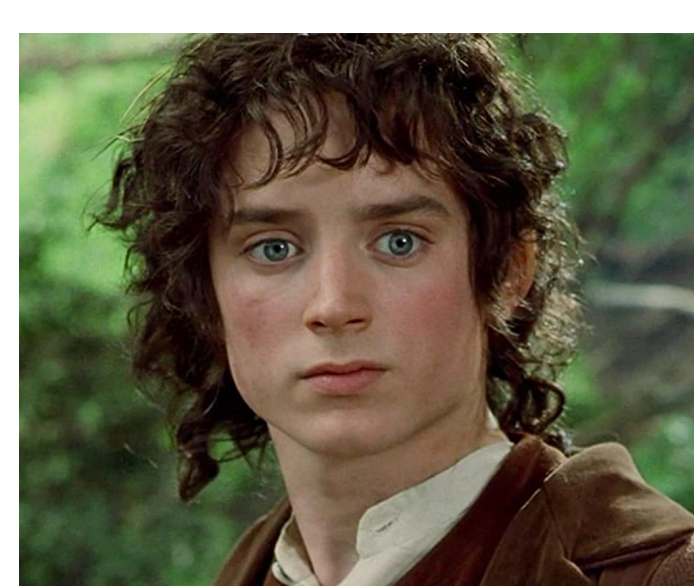
- A leitmotif (*leading motif*) is a recurring theme or „motif“
- Carries symbolic significance in all forms of art, e.g. literature, movies, and music
- Humans associate leitmotifs with meaning, which enhances narrative cohesion and establishes emotional connections with the audience

Motivation: Leitmotifs in Soundtracks

- Genre often using leitmotifs are soundtracks, for instance in compositions of Hans Zimmer, Howard Shore or John Williams.



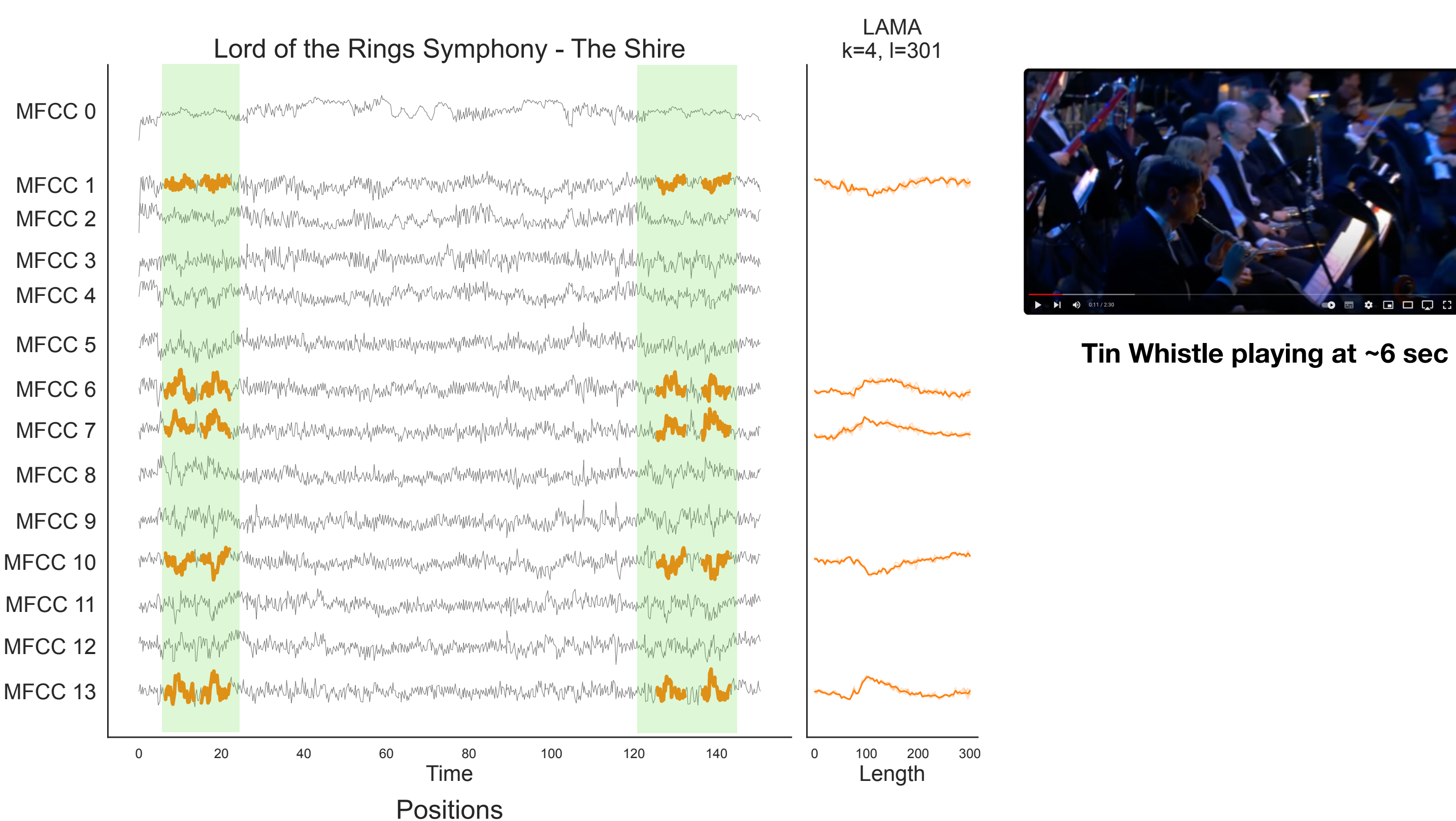
Star Wars - Imperial March



Lord of the Rings - The Shire

Example: Lord of the Rings – The Shire

- The Shire was arranged by Howard Shore for Lord of the Rings
- It opens and ends with the **Hobbits’ leitmotif**, played by a solo tin whistle
- The leitmotif manifests in a distinct pattern in **several, but not all frequencies** (aka dimensions)

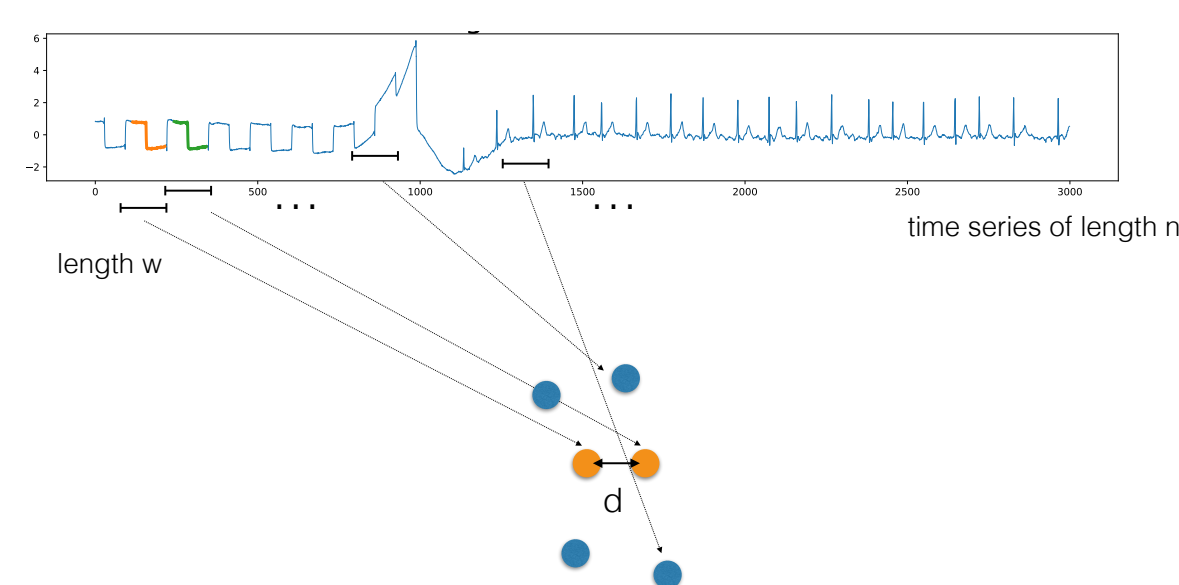


Challenging Problem: Subdimensional Motif Set Discovery

- Discovery of leitmotifs is interesting in motion capture, sensor or stock data, too
- It is an unsupervised and mathematically ill-defined problem of high complexity
- Leitmotifs have:
 1. Unknown length l ,
 2. Occur an unknown number of times k , and
 3. Occur in an unknown number of sub-dimensions f

High-Dim Vector-Space

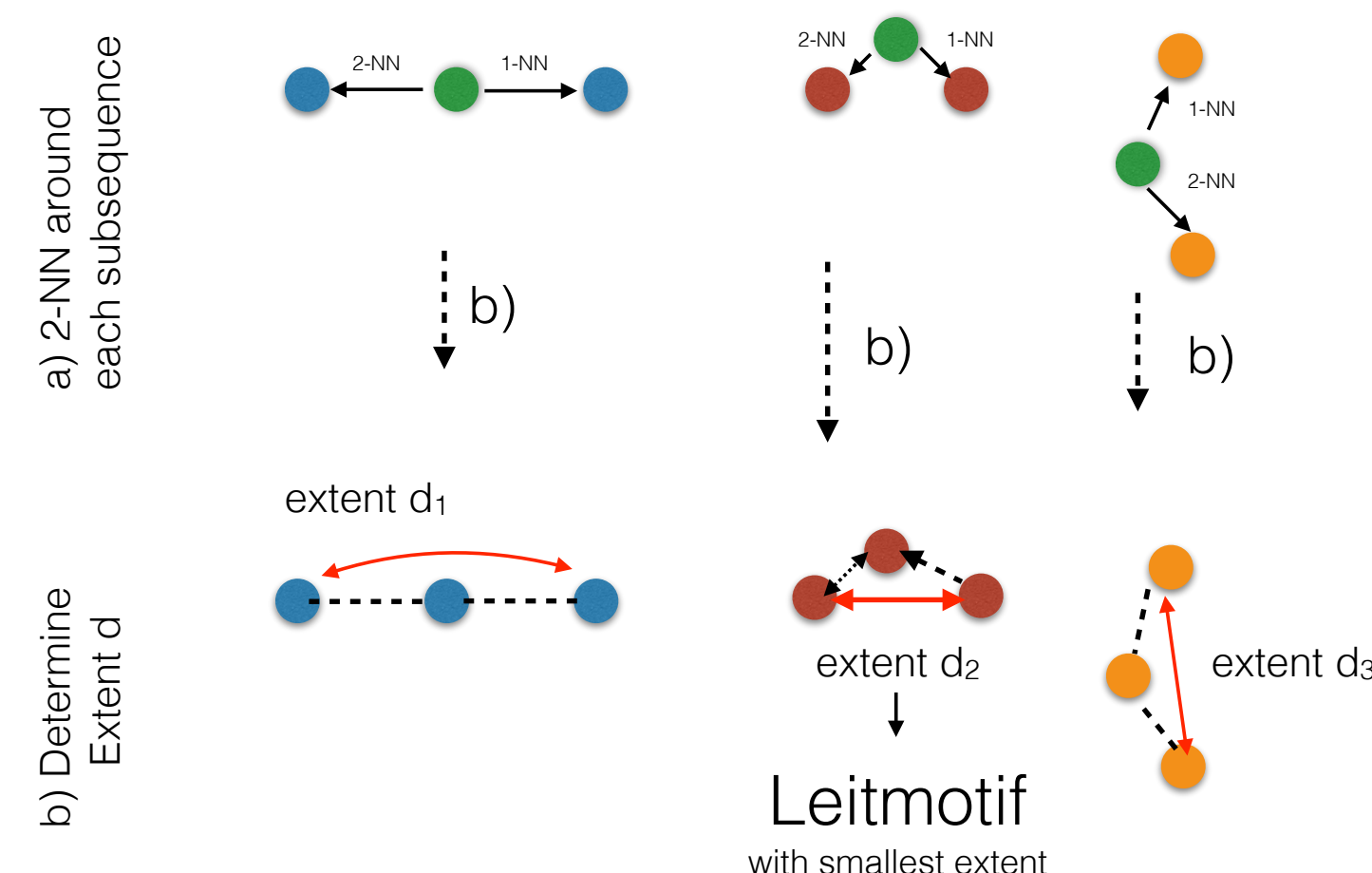
- We extract w -length subsequences, and interpret these as points in w -dim space
- Similar sets of points are motif set candidates



Motiflets – Univariate

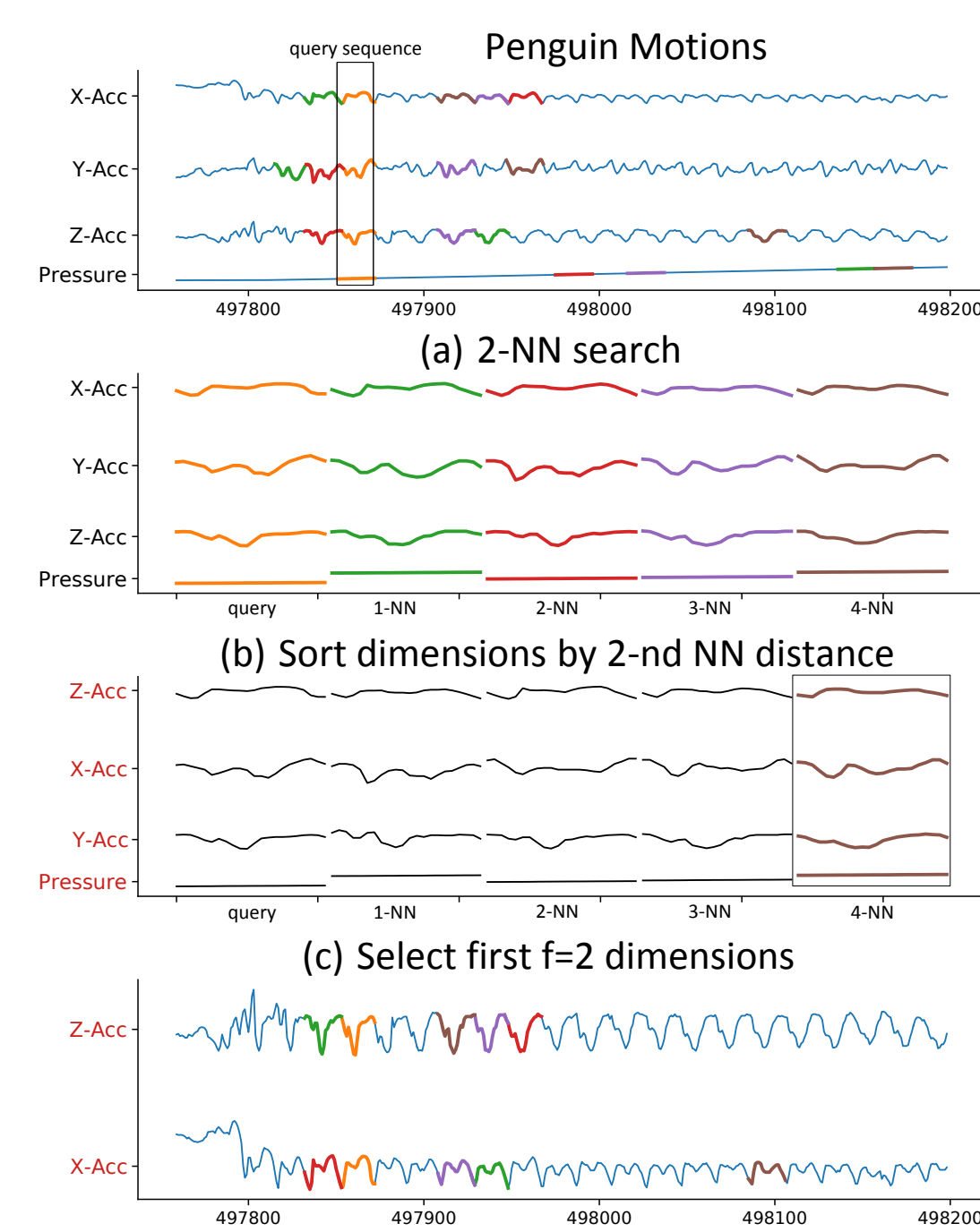
Designed to find k -frequent motifs using a single dimension

1. k -NN search around candidate subsequence
2. Compute maximum over pairwise distances (“extent”)
3. Minimizes extent over all candidates

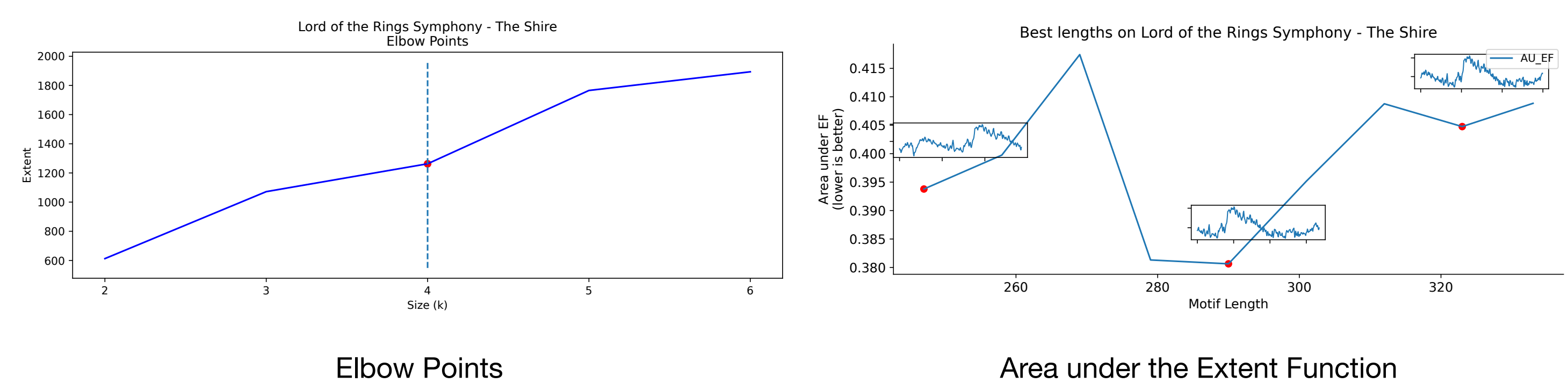


LAMA – Subdimensional Motif Set Discovery

- Goal: Search k -freq motif sets of length l in latent f -dim subspace with minimal extent (NP-hard)
- First, for each subsequence s :
 - (a) perform k -NN search along each dimension
 - (b) sort dimensions by k^{th} NN distance: $dist_{kNN}$
 - (c) take f smallest $dist_{kNN}$ values
 - (d) use k -NN offsets from dimension with smallest $dist_{kNN}$
 - (e) compute candidate extent as sum over f dims
- Finally, minimize extent over all f -dim candidates

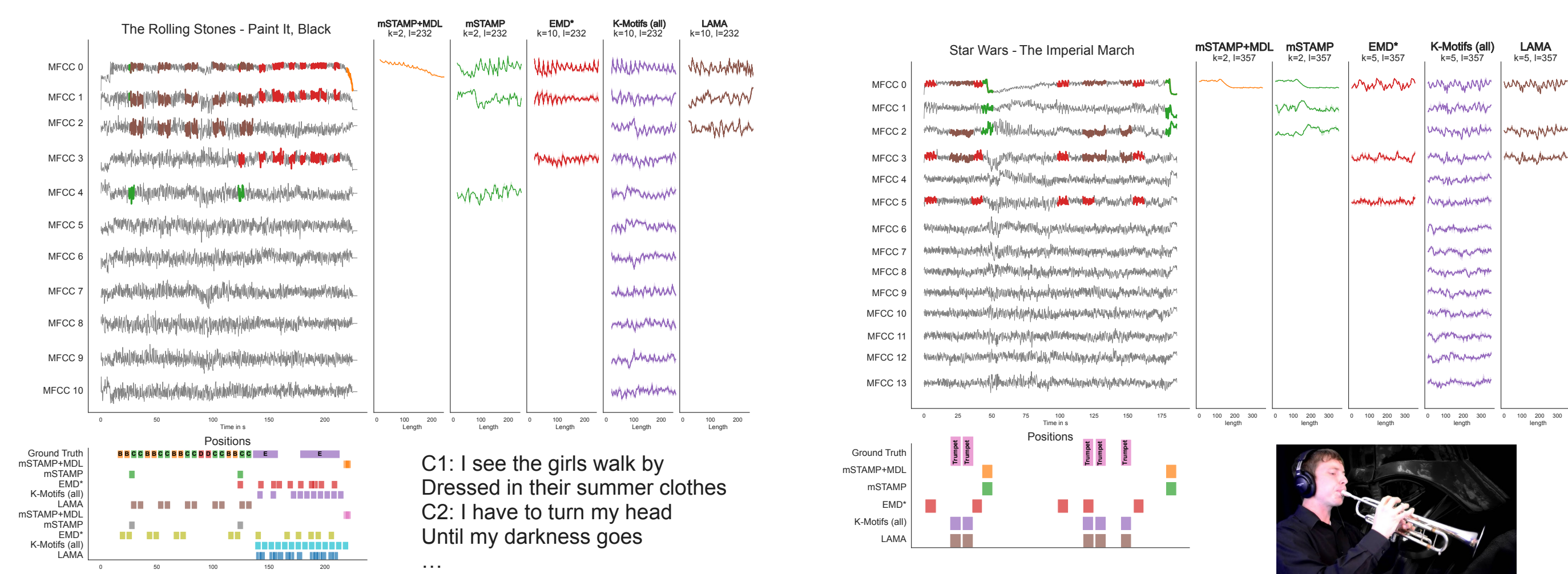


Learning l and k (see paper)



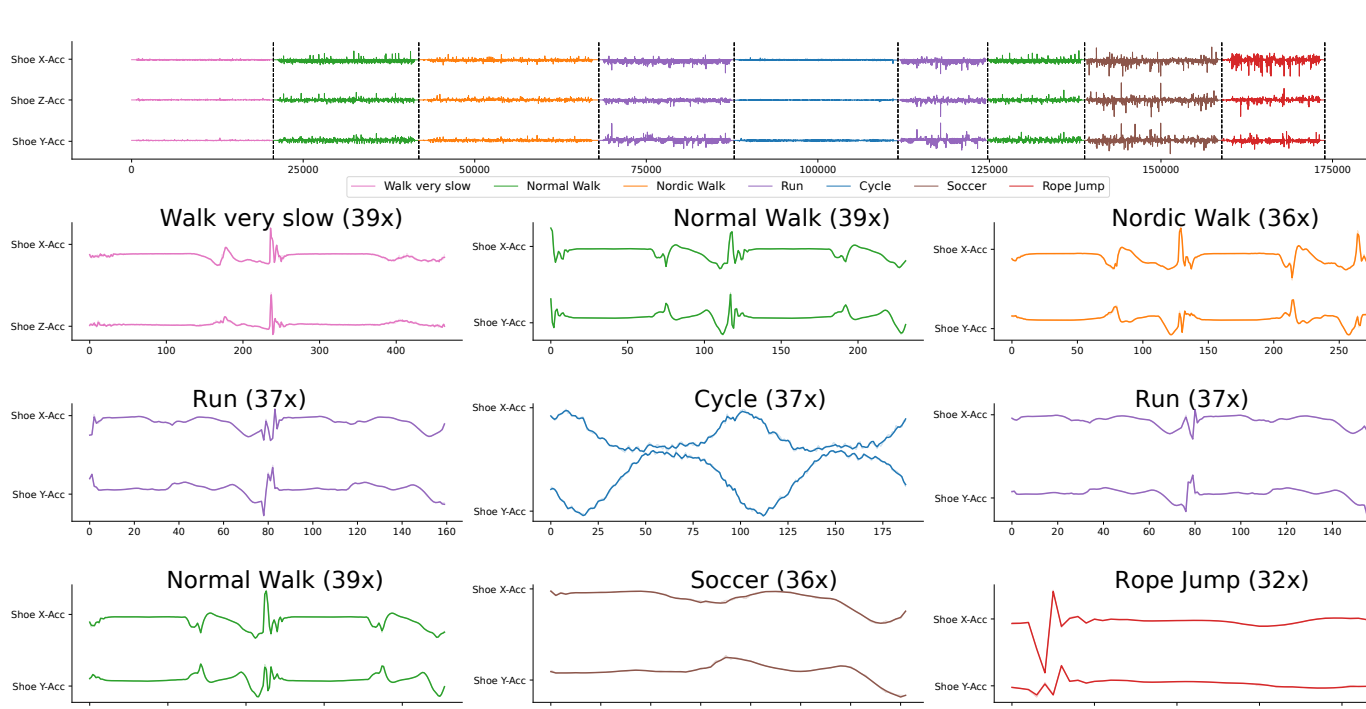
Evaluation

- We evaluated LAMA on 14 real-world use cases against 4 competitors

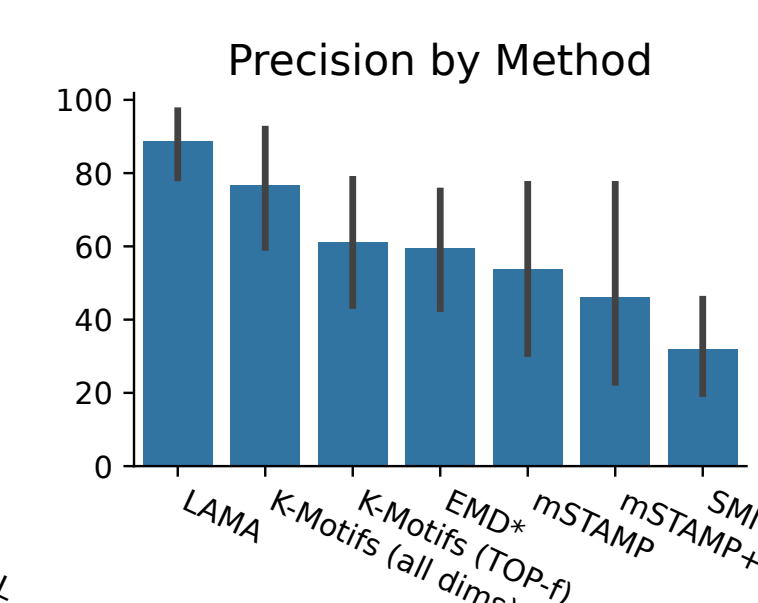
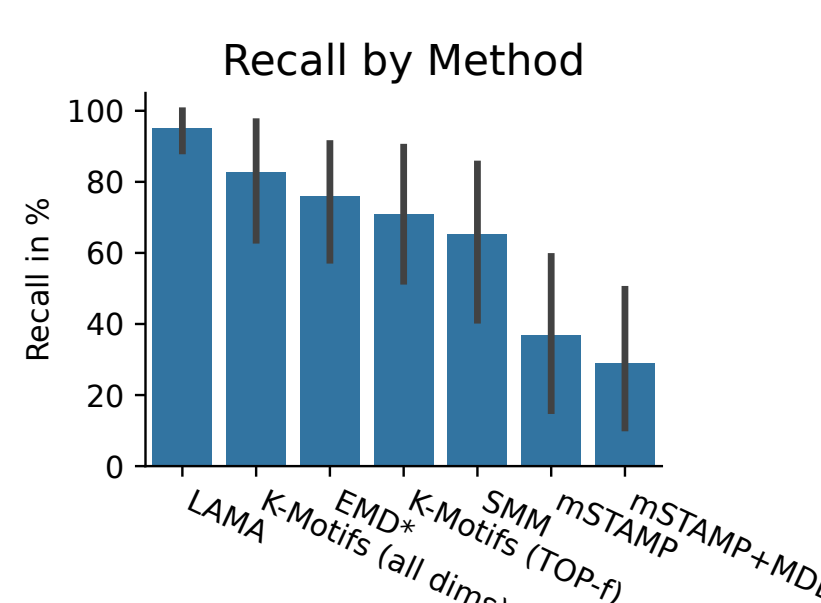


- Music: Leitmotif is the C part with 10 repeats

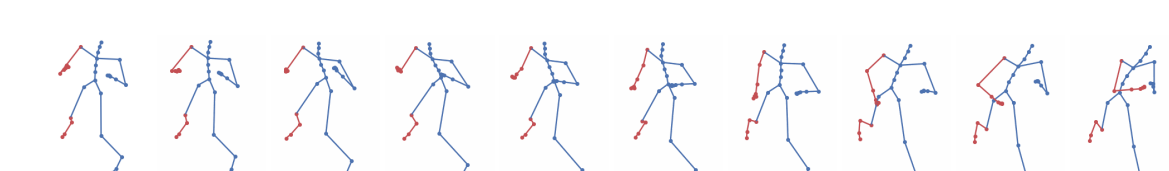
- Soundtrack: Leitmotif is the trumpets playing



- Sensor Data: Leitmotifs are different walking styles



- Motion Capture: Leitmotif is a punching motion



Conclusion

- Our experimental evaluation of 14 distinct real-life data sets shows that LAMA shows superior performance in detecting meaningful patterns
- Our findings demonstrate that summarizing a time series through its leitmotifs offers a profound understanding of underlying patterns

[1] Schäfer, Patrick, and Ulf Leser. "Motiflets: Simple and Accurate Detection of Motifs in Time Series." Proceedings of the VLDB Endowment 16.4

[2] Schäfer, Patrick, and Ulf Leser. "Discovering Leitmotifs in Multidimensional Time Series." Proceedings of the VLDB Endowment 18.2